

Variables for our project

<https://www.dol.gov/agencies/ebsa/about-ebsa/our-activities/public-disclosure/foia/form-5500-datasets>

Inflation as a control - year-to-year growth -

<https://www.macrotrends.net/global-metrics/countries/USA/united-states/inflation-rate-cpi>

S&P 500 year-to-year growth rate -

<https://www.macrotrends.net/2526/sp-500-historical-annual-returns>

Federal reserve Interest rate -

<https://www.kaggle.com/datasets/federalreserve/interest-rates>

Policy Analysis of Affordable Health Care Act (ACA)

Trying to see if after implementation of ACA people received more premiums (did the act work for the average American).

[Data](#) from US Department of Labour

Methods

Difference in difference [40 states which did adopt ACA vs 10 states which did not adopt it](#)

- Parallel assumption issue (trial and error fix)

or some ML method because of the large amount of data

Y-var (Premiums per capita)

WLFR_TOT_CHARGES_PAID_AMT: - non experience-rated contracts premiums amount received

or

WLFR_PREMIUM_RCVD_AMT: - experience rate plan premiums received

Divided by amount of RETIREES in the pension plan

Controls / X

State indicator for Difference in Difference (Not all states have ACA)

Year indicator (Before / After ACA)

Inflation yearly

S&P 500 year-year growth rate for market indicator

Federal Reserve interest rates yearly

Variables

ID (plan Identification number)

INS_PRSN_COVERED_EOY_CNT (Insured Persons Covered End of Year Count)

- Units: Number of people
- Descriptive Statistics for INS_PRSN_COVERED_EOY_CNT:
 - Average Statistics for INS_PRSN_COVERED_EOY_CNT across all sheets:
 - Mean of Means: 1333.67
 - Mean of Standard Deviations: 42041.70
 - Overall Minimum: 0.00
 - Overall Maximum: 9999999.00

WLFR_PREMIUM_RCVD_AMT (Welfare Premium Received Amount. It represents the total premiums received by the insurance company for the coverage provided during the policy year. This is only for experience-rated contracts.)

- Units: US Dollars
- Descriptive Stats
 - Average Statistics for WLFR_PREMIUM_RCVD_AMT across all sheets:
 - Mean of Means: 2394116.44
 - Mean of Standard Deviations: 937505906.19
 - Overall Minimum: -130923183.00
 - Overall Maximum: 8701261226322.00

WLFR_TOT_CHARGES_PAID_AMT (Welfare Total Charges Paid Amount. It represents the total amount of charges paid by the insurance company for claims during the policy year. This is for non-experience rated contracts.)

- Units: US Dollars
- Descriptive Statistics
 - Average Statistics for WLFR_TOT_CHARGES_PAID_AMT across all sheets:
 - Mean of Means: 2900620.40
 - Mean of Standard Deviations: 1250374247.52
 - Overall Minimum: -50876783.00
 - Overall Maximum: 7708480105986.00

SPONS_DFE_STATE (This variable represents the plan sponsor / Direct Filing Entity (DFE). It indicates the state where the plan sponsor or DFE is located.)

- Units: two letter state ID's

ADMIN_STATE (This represents the state of the plan administrator. It indicates the state where the plan's administrative operations are primarily conducted).

- Units: two letter state ID's

WLFR_PREMIUM_RCVD_AMT_PER_CAPITA (Dividing WLFR_PREMIUM_RCVD_AMT by INS_PRSN_COVERED_EOY_CNT)

- Units: US Dollars
- Descriptive Stats
 - Average Statistics for WLFR_PREMIUM_RCVD_AMT_PER_CAPITA across all sheets:
 - Mean of Means: 21864.86
 - Mean of Standard Deviations: 13302903.79
 - Overall Minimum: -178856.81
 - Overall Maximum: 204126900569.05

WLFR_TOT_CHARGES_PAID_AMT_PER_CAPITA (Dividing WLFR_TOT_CHARGES_PAID_AMT by INS_PRSN_COVERED_EOY_CNT).

- Units: US Dollars
- Descriptive Stats
 - Mean of Means: 33723.96
 - Mean of Standard Deviations: 16920728.97
 - Overall Minimum: -4771105.50
 - Overall Maximum: 211823456790.11

Control Variables

Inflation

- Inflation_rate, Inflation_rate_annual_change
 - Units: Percentage

Market Indicator (S&P 500 Yearly Growth)

- Units: Percentage

Federal Reserve Interest Rates

- Unit: Interest Rate Percentage

Inflation has two variable names - Inflation_rate and Inflation_rate_annual_change, with time being called Year

S&P Yearly Growth - has the variable S&P Yearly Growth, and date as time

Federal Reserve Interest Rates has Year, Month and Day for time and Effective Federal Funds Rate for Interest Rates.

Statistics for Inflation:

Mean: 0.0255

Standard Deviation: 0.0163

Minimum: -0.0036

Maximum: 0.0800

Statistics for Inflation_annual_change:

Mean: 0.0010

Standard Deviation: 0.0176

Minimum: -0.0419

Maximum: 0.0346

Statistics for S&P_Yearly_Growth:

Mean: 7.1780

Standard Deviation: 17.7607

Minimum: -38.4900

Maximum: 29.6000

Statistics for Federal_Funds_Rate_Average_Yield:

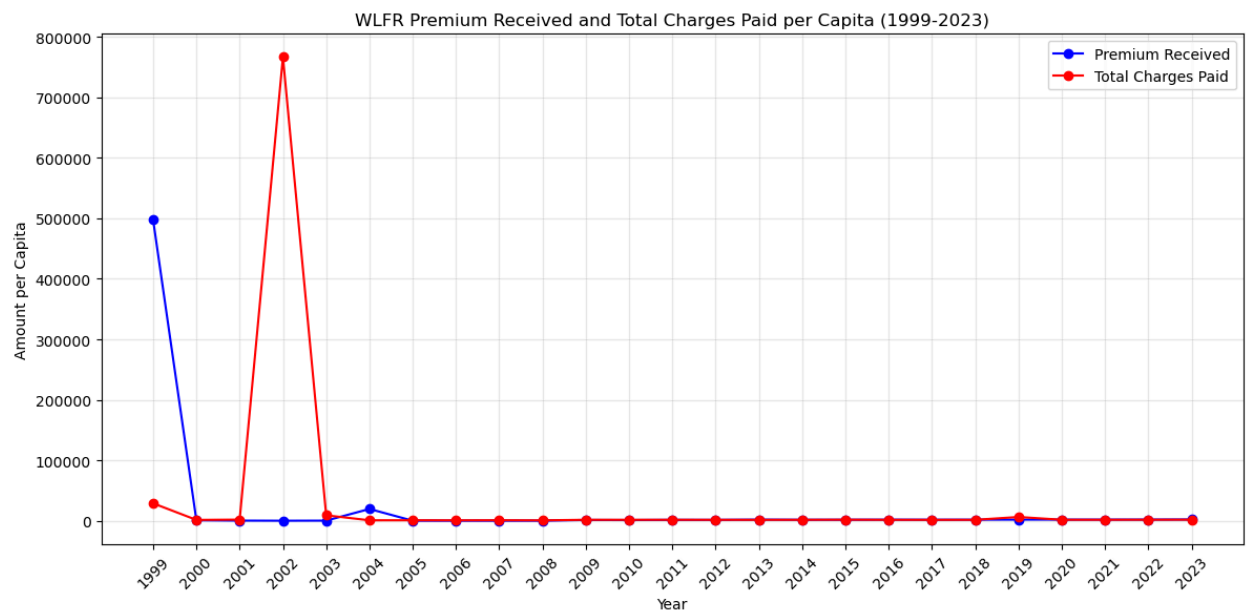
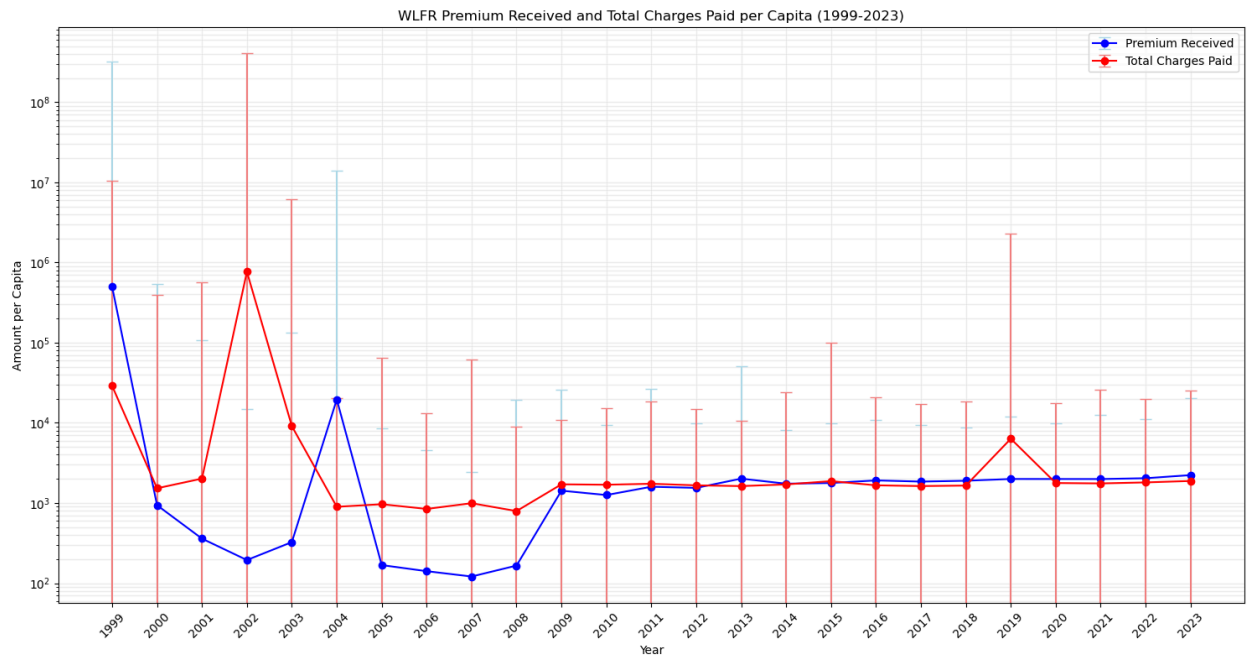
Mean: 0.0191

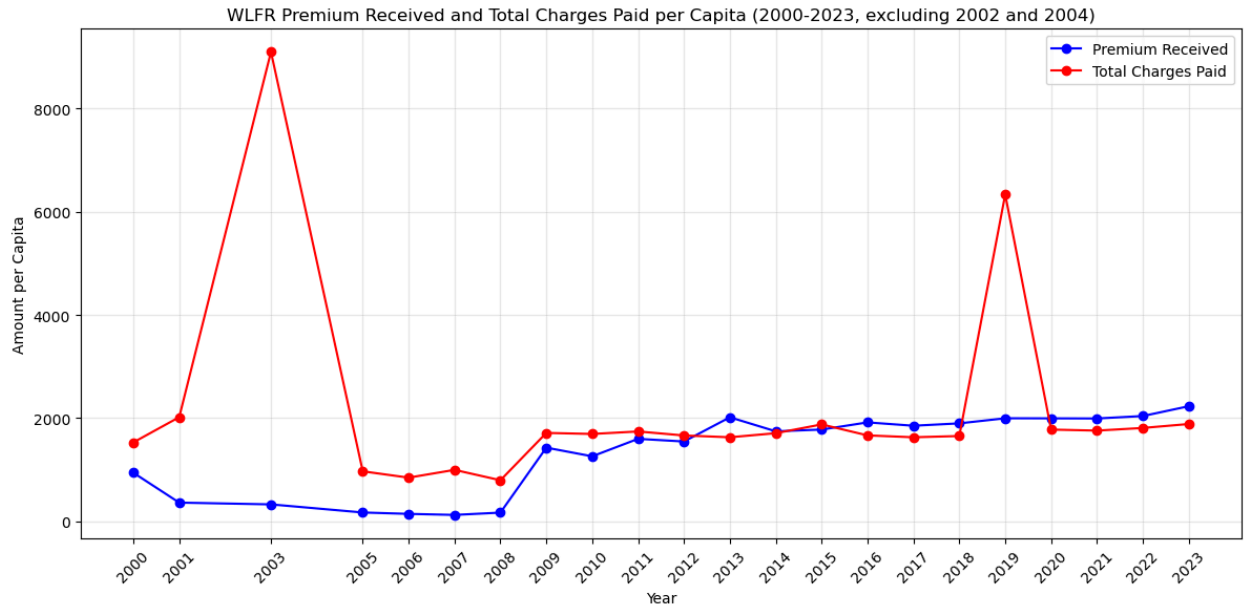
Standard Deviation: 0.0199

Minimum: 0.0008

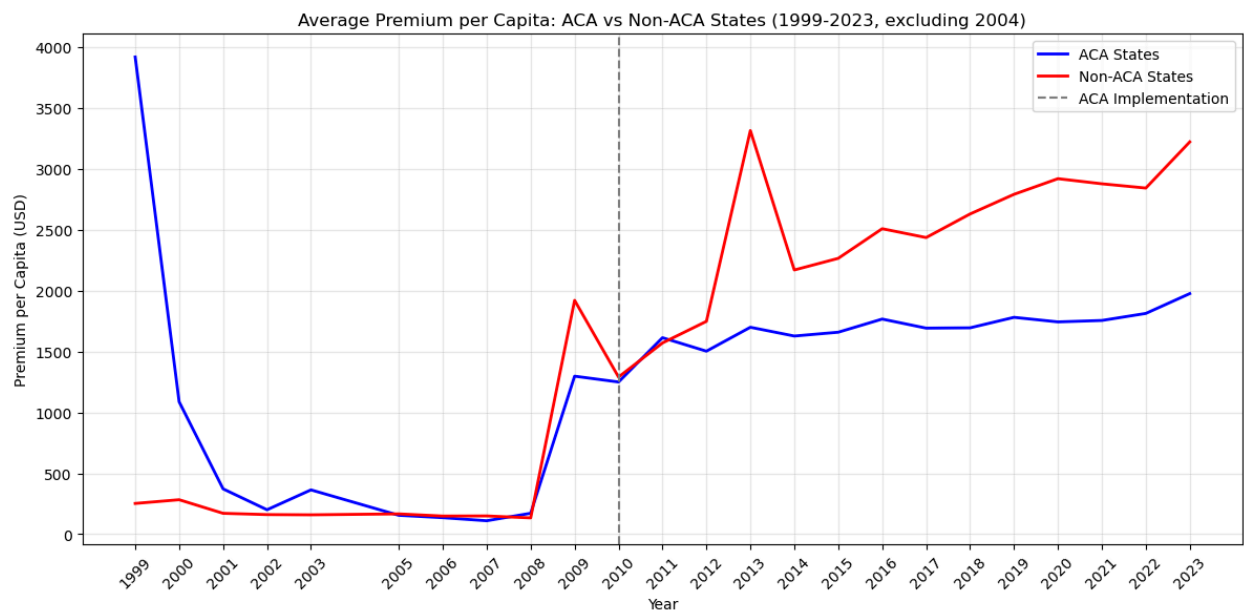
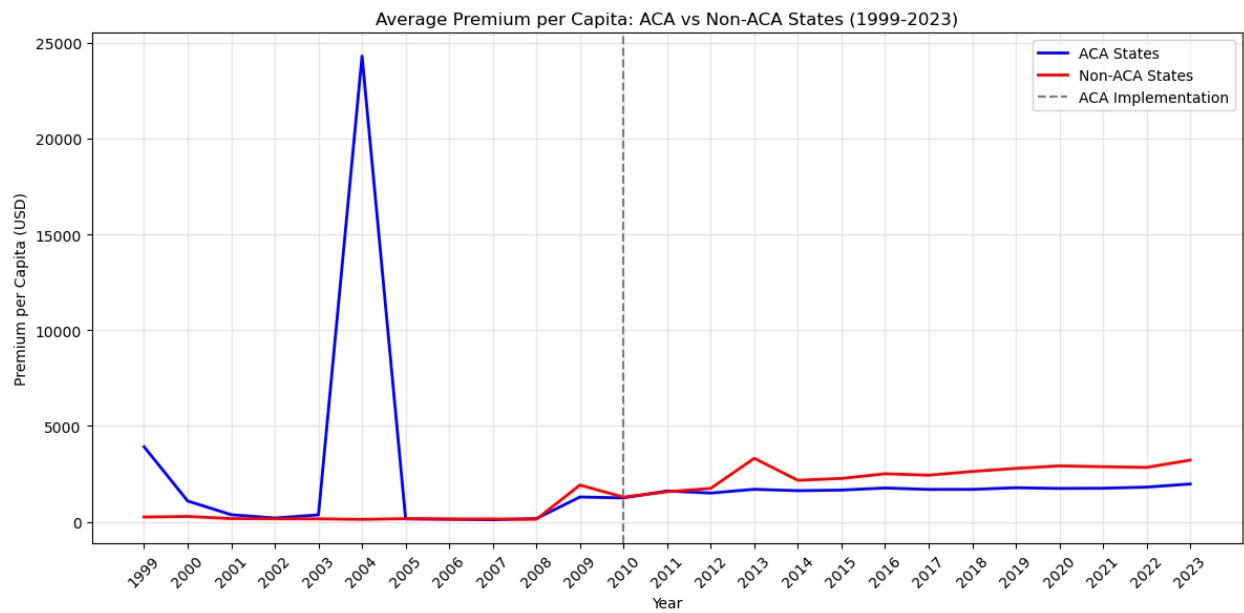
Maximum: 0.0624

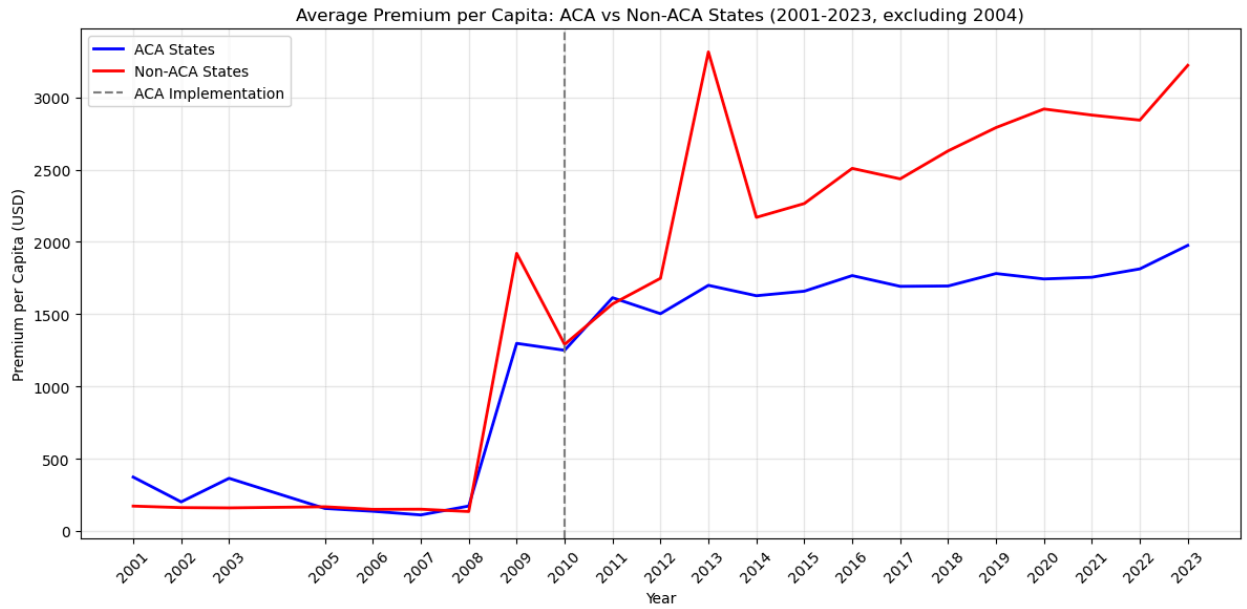
Our Y Variable over time



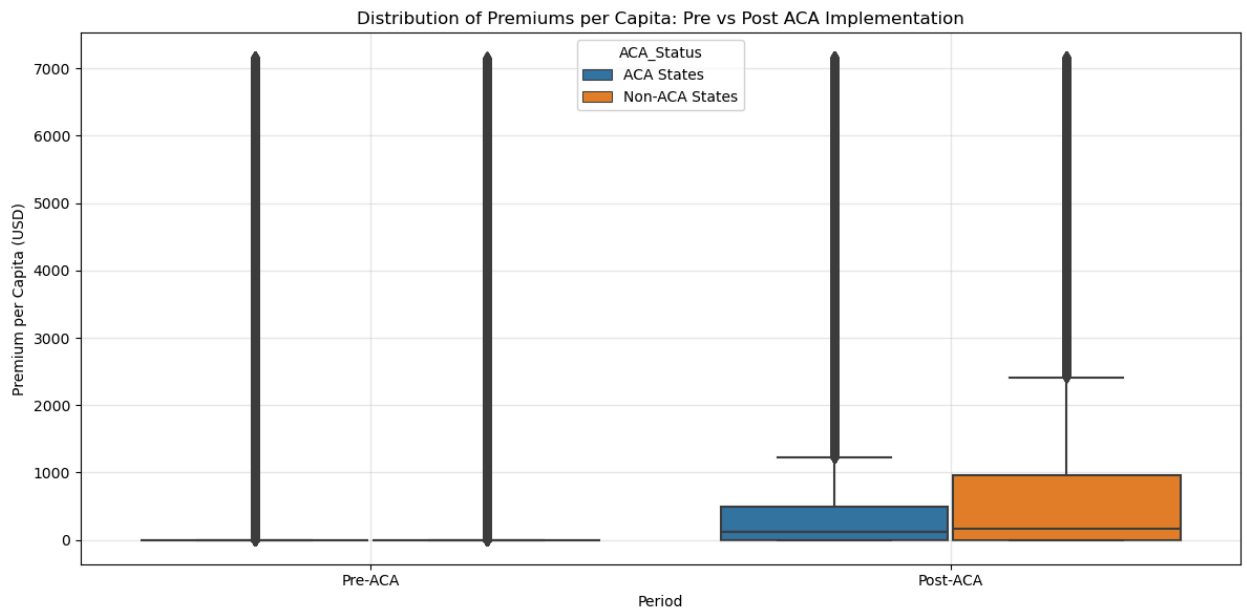


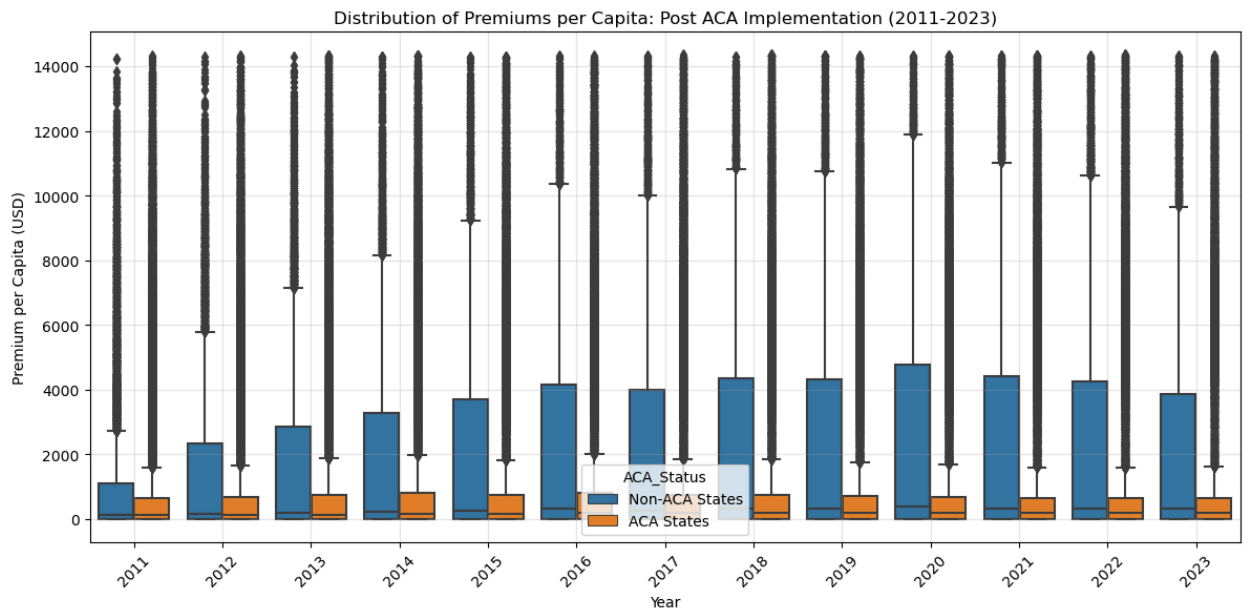
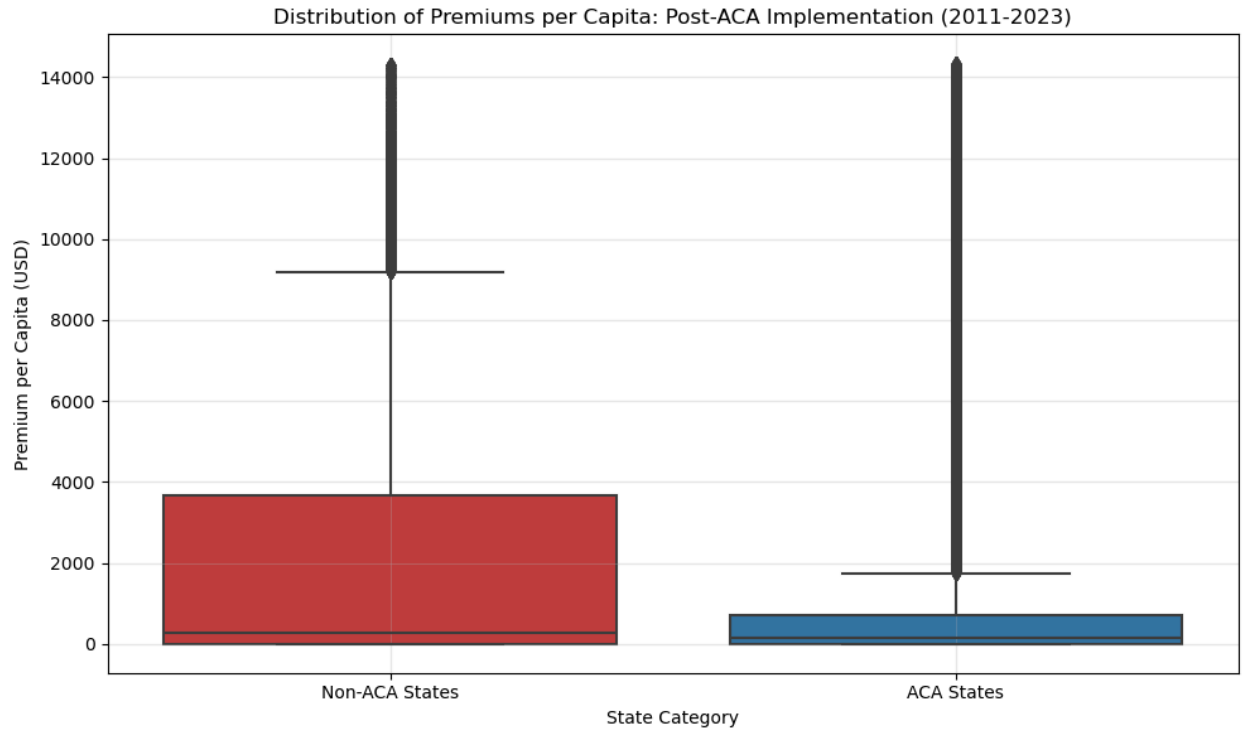
Parallel Assumption Graphic





Distribution of premiums per capita





Methodology

The method we are going to use for our analysis is a difference-in-difference (DiD) regression to find the causal effect of the Affordable Care Act (ACA) on insurance premiums per person. We have decided on this approach for three reasons.

- The implementation of the ACA creates a natural experiment where 40 states implemented it and 10 did not.

- We had data from 1999-2023 where ACA was implemented in 2010 giving us long term post and pre-treatment data.
- The implementation is also clear with the ACA taking place in 2010.

The regression equation we will use is, we will do two regression one for experience based plans and one for non experience-based plans.

$$Y_{ist} = \beta_0 + \beta_1 ACA_s + \beta_2 Post_t + \beta_3 (ACA_s * Post_t) + \gamma Controls_t + \varepsilon_{ist}$$

Y_{ist} - represents premiums per capita for either experience based plans or non experience-based plans. i is plan, state is s and time is t.

ACA_s - dummy for if the state adopted the ACA

$Post_t$ - is a dummy for post-2010 periods

$Controls$ - represents a vector of all of our controls (inflation, Market indicator (S&P 500 growth), Federal Interest Rates)

ε_{ist} - error term

In order to apply DiD we need to have parallel trends assumption which in this context would mean that that treatment and control states would have followed similar trends in the absence of treatment (the ACA). We have passed this check with the graph above. For this parallel trend to hold we had to omit 1999,2000 and 2004 as they had very inconsistent data. Along with this to deal with outlier we omitted 1% of outliers which was apparent from our maximums and minimums from the summary statistics of our variables.

The model includes macroeconomic controls to account for factors that might affect premium rates independently of the ACA:

- Annual inflation rate
- S&P 500 year-over-year growth rate
- Federal Reserve interest rates

These controls help isolate the ACA's effect from broader economic conditions that could influence insurance markets and premiums from the plans.

ROBUSTNESS CHECKS MAYBE

Regressions

First one (no controls, no negative values, and removed extreme outliers (99th percentile)).

DiD Results for Experience-Rated Plans (Premium):

```
-----
      Coefficient Std Error   t-stat   p-value CI Lower \
const      6.870518  0.019320 355.623541  0.000000e+00 6.832653
ACA_State  -0.209368  0.022583 -9.271187  1.840851e-20 -0.253629
Post_ACA   0.867870  0.023381 37.118801  1.397625e-301 0.822044
ACA_x_Post -0.092507  0.029108 -3.178061  1.482635e-03 -0.149558
```

```
      CI Upper
const      6.908384
ACA_State  -0.165107
Post_ACA   0.913695
ACA_x_Post -0.035456
R-squared: 0.0514
Number of observations: 67821.0
```

Diagnostic Tests:
F-statistic: 1474.5913
Prob(F-statistic): 0.0000

DiD Results for Non-Experience-Rated Plans (Charges):

```
-----
      Coefficient Std Error   t-stat   p-value CI Lower \
const      4.666593  0.021077 221.401946  0.000000e+00 4.625282
ACA_State   0.680978  0.024206 28.132448  3.929498e-174 0.633535
Post_ACA    0.302515  0.029318 10.318237  5.828373e-25 0.245052
ACA_x_Post  -0.019680  0.034679 -0.567496  5.703770e-01 -0.087649
```

```
      CI Upper
const      4.707904
ACA_State   0.728421
Post_ACA    0.359978
ACA_x_Post  0.048289
R-squared: 0.0246
Number of observations: 67821.0
```

Diagnostic Tests:
F-statistic: 573.7271
Prob(F-statistic): 0.0000

Regression results (Controls + log transformed dependant variable)

Data Shape: (55671, 12)

Missing Values:

Year	0
ACA_State	0
Post_ACA	0
ACA_x_Post	0
WLFR_PREMIUM_RCVD_AMT_PER_CAPITA	0
WLFR_TOT_CHARGES_PAID_AMT_PER_CAPITA	0
Inflation	0
Inflation_annual_change	0
S&P_Yearly_Growth	0
Federal_Funds_Rate_Average_Yield	0
Log_Premium	0
Log_Charges	0

dtype: int64

Summary Statistics:

	Year	ACA_State	Post_ACA	ACA_x_Post \
count	55671.000000	55671.000000	55671.000000	55671.000000
mean	2010.595337	0.691114	0.480555	0.307377
std	6.822652	0.462038	0.499626	0.461411
min	2001.000000	0.000000	0.000000	0.000000
25%	2005.000000	0.000000	0.000000	0.000000
50%	2010.000000	1.000000	0.000000	0.000000
75%	2016.000000	1.000000	1.000000	1.000000
max	2023.000000	1.000000	1.000000	1.000000

	WLFR_PREMIUM_RCVD_AMT_PER_CAPITA	WLFR_TOT_CHARGES_PAID_AMT_PER_CAPITA \
count	5.567100e+04	5.567100e+04
mean	4.367475e+03	6.964918e+03
std	6.116263e+04	1.384253e+06
min	2.700626e-05	6.157635e-04
25%	4.605129e+02	5.249131e+01
50%	2.801627e+03	2.803864e+02
75%	5.135636e+03	9.022021e+02
max	8.981869e+06	3.266040e+08

	Inflation	Inflation_annual_change	S&P_Yearly_Growth \
count	55671.000000	55671.000000	55671.000000
mean	0.024878	-0.000260	5.475276

std	0.015744	0.016917	18.366767
min	-0.003600	-0.041900	-38.490000
25%	0.015900	-0.006300	-6.240000
50%	0.022700	-0.001700	9.540000
75%	0.032300	0.008700	23.450000
max	0.080000	0.034600	29.600000

	Federal_Funds_Rate_Average_Yield	Log_Premium	Log_Charges
count	55671.000000	55671.000000	55671.000000
mean	0.017800	7.243980	5.330767
std	0.017307	1.848912	2.077695
min	0.000800	-10.519442	-7.392648
25%	0.001600	6.132341	3.960648
50%	0.011300	7.937955	5.636168
75%	0.032200	8.543959	6.804839
max	0.050300	16.010719	19.604259

Results for Premium Received (Experience-Rated Plans):

	Coefficient	Std Error	t-stat	\
const	7.198970	0.025213	285.530316	
ACA_State	-0.335350	0.025148	-13.335002	
Post_ACA	0.472023	0.026756	17.641860	
ACA_x_Post	0.082578	0.031589	2.614118	
Inflation	4.977947	0.779106	6.389307	
Inflation_annual_change	-3.849609	0.643241	-5.984708	
S&P_Yearly_Growth	0.003403	0.000469	7.261170	
Federal_Funds_Rate_Average_Yield	-6.680602	0.696370	-9.593467	

	p-value	CI Lower	CI Upper
const	0.000000e+00	7.149555	7.248386
ACA_State	1.448514e-40	-0.384639	-0.286061
Post_ACA	1.175273e-69	0.419582	0.524463
ACA_x_Post	8.945809e-03	0.020664	0.144492
Inflation	1.666395e-10	3.450927	6.504966
Inflation_annual_change	2.167781e-09	-5.110338	-2.588880
S&P_Yearly_Growth	3.837555e-13	0.002485	0.004322
Federal_Funds_Rate_Average_Yield	8.517392e-22	-8.045463	-5.315742

R-squared: 0.04362655936174786

Adjusted R-squared: 0.04350628891127872

F-statistic: 380.7084503020798

Prob (F-statistic): 0.0

Results for Total Charges Paid (Non-Experience-Rated Plans):

	Coefficient	Std Error	t-stat	\
const	4.251346	0.030465	139.547874	
ACA_State	0.754043	0.028369	26.579469	
Post_ACA	0.591931	0.034877	16.971817	
ACA_x_Post	-0.153902	0.038371	-4.010838	
Inflation	6.979188	0.930124	7.503505	
Inflation_annual_change	-1.813180	0.745825	-2.431105	
S&P_Yearly_Growth	0.000173	0.000530	0.325961	
Federal_Funds_Rate_Average_Yield	8.207756	0.765621	10.720390	

	p-value	CI Lower	CI Upper
const	0.000000e+00	4.191636	4.311057
ACA_State	1.172660e-155	0.698440	0.809646
Post_ACA	1.327594e-64	0.523573	0.660289
ACA_x_Post	6.050368e-05	-0.229108	-0.078695
Inflation	6.213377e-14	5.156179	8.802197
Inflation_annual_change	1.505284e-02	-3.274970	-0.351389
S&P_Yearly_Growth	7.444541e-01	-0.000866	0.001211
Federal_Funds_Rate_Average_Yield	8.165832e-27	6.707166	9.708345

R-squared: 0.03690402702749962

Adjusted R-squared: 0.036782911172967836

F-statistic: 297.79940862116376

Prob (F-statistic): 0.0

Interpretation of Regression

Our regression equation is:

$$Y_{ist} = \beta_0 + \beta_1 ACA_s + \beta_2 Post_t + \beta_3 (ACA_s * Post_t) + \gamma Controls_t + \varepsilon_{ist}$$

It is important to note that we ran this equation twice: once for Experience-Rated plans and once for non-experience rated plans.

The main coefficients we are interested in is β_3 which represents the effect of the ACA on premiums per person in states that adopted it, compared to those that didn't, after implementation. It is also worth analyzing the effect of β_1 . This coefficient represents the difference between ACA adopting states and non ACA adopting states.

We ran two regressions so far, one without the vector of control variables and one with them present.

The first regression (without our control variable vector):

Experience Rated Plans:

$$ACA_s (\beta_1) -0.209368$$

- ACA States had 20.94% lower premiums per person than non ACA states before the implementation of the ACA.
- Significant at p-value < 0.05

$$ACA_s * Post_t (\beta_3) -0.092507$$

- After the ACA implementation premiums per person in ACA states decreased by an additional 9.25% compared to non-ACA states.
- The 95% confidence interval is [-0.149558, -0.035456] meaning that we can assume a negative effect.
- Significant at p-value < 0.05

Non-Experience-Rated Plans

$$ACA_s (\beta_1) 0.680978$$

- ACA states had 68.10% higher premiums per person than non-ACA states before the ACA was implemented.
- Significant at p-value < 0.05

$$ACA_s * Post_t (\beta_3) -0.019680$$

- Not statistically significant (not distinguishable from 0 at p-value < 0.05)
- 1.97% decrease in premiums per person for ACA states after the implementation of the ACA.

Overall: For Experience Rated Plans there is evidence that premiums per person significantly decreased and for Non-Experience-Rated Plans there seems to have been no effect from the implementation of the ACA.

The second regression (with our control variable vector):

Experience Rated Plans:

$ACA_s (\beta_1) -0.335350$

- ACA states had 33.54% lower premiums per person than non-ACA states before the implementation of the ACA.
- Statistically significant and 95% confidence interval is [-0.384639, -0.286061] suggesting a significant negative effect.

$ACA_s * Post_t (\beta_3) 0.082578$

- After the ACA implementation premiums per person in ACA states increased by 8.26%.
- Statistically significant with the 95% confidence interval: [0.020664, 0.144492] suggesting a significant positive effect.

Non-Experience-Rated Plans

$ACA_s (\beta_1) 0.754043$

- ACA states had 75.40% higher premiums per person than non-ACA states before the ACA implementation.
- The 95% confidence interval is [0.698440, 0.809646] suggesting a very significant positive effect.

$ACA_s * Post_t (\beta_3) -0.153902$

- After the ACA implementation premiums per person in ACA states decreased by 15.39% compared to non ACA states.
- Statistically significant with the 95% confidence interval being [-0.229108, -0.078695].

Overall: For experience rated plans the ACA increased premiums per person while for Non-Experience-Rated Plans it decreased them, both results are significant.

Appendix: Detailed Variables across sheets

INS_PRSN_COVERED_EOY_CNT

Average Statistics for INS_PRSN_COVERED_EOY_CNT across all sheets:

Mean of Means: 1333.67

Mean of Standard Deviations: 42041.70

Overall Minimum: 0.00

Overall Maximum: 9999999.00

Statistics for each sheet:

Sheet: Year_1999

Mean: 1388.66

Standard Deviation: 68332.65

Minimum: 0.00

Maximum: 9869819.00

Sheet: Year_2000

Mean: 1630.04

Standard Deviation: 72879.79

Minimum: 0.00

Maximum: 9529314.00

Sheet: Year_2001

Mean: 1339.90

Standard Deviation: 60689.85

Minimum: 0.00

Maximum: 9700209.00

Sheet: Year_2002

Mean: 774.66

Standard Deviation: 30152.49

Minimum: 0.00

Maximum: 9082060.00

Sheet: Year_2003

Mean: 793.28

Standard Deviation: 42871.13
Minimum: 0.00
Maximum: 9999999.00

Sheet: Year_2004
Mean: 881.82
Standard Deviation: 38809.28
Minimum: 0.00
Maximum: 9851283.00

Sheet: Year_2005
Mean: 944.31
Standard Deviation: 41774.22
Minimum: 0.00
Maximum: 8335582.00

Sheet: Year_2006
Mean: 1053.30
Standard Deviation: 47255.47
Minimum: 0.00
Maximum: 9305115.00

Sheet: Year_2007
Mean: 1470.90
Standard Deviation: 68859.05
Minimum: 0.00
Maximum: 8870611.00

Sheet: Year_2008
Mean: 7339.91
Standard Deviation: 192799.09
Minimum: 0.00
Maximum: 9902050.00

Sheet: Year_2009
Mean: 915.33
Standard Deviation: 25566.35
Minimum: 0.00
Maximum: 9122400.00

Sheet: Year_2010
Mean: 1054.79
Standard Deviation: 35088.26
Minimum: 0.00

Maximum: 9782251.00

Sheet: Year_2011

Mean: 1041.82

Standard Deviation: 30644.85

Minimum: 0.00

Maximum: 9708140.00

Sheet: Year_2012

Mean: 1167.65

Standard Deviation: 44624.39

Minimum: 0.00

Maximum: 9906634.00

Sheet: Year_2013

Mean: 983.74

Standard Deviation: 25237.10

Minimum: 0.00

Maximum: 9416328.00

Sheet: Year_2014

Mean: 998.37

Standard Deviation: 15940.98

Minimum: 0.00

Maximum: 3332552.00

Sheet: Year_2015

Mean: 1078.22

Standard Deviation: 28286.42

Minimum: 0.00

Maximum: 8039970.00

Sheet: Year_2016

Mean: 998.98

Standard Deviation: 20000.58

Minimum: 0.00

Maximum: 8083571.00

Sheet: Year_2017

Mean: 1043.85

Standard Deviation: 27478.87

Minimum: 0.00

Maximum: 7313708.00

Sheet: Year_2018
Mean: 1076.05
Standard Deviation: 28779.25
Minimum: 0.00
Maximum: 8982947.00

Sheet: Year_2019
Mean: 993.73
Standard Deviation: 19792.82
Minimum: 0.00
Maximum: 7080959.00

Sheet: Year_2020
Mean: 1008.59
Standard Deviation: 20235.26
Minimum: 0.00
Maximum: 8010120.00

Sheet: Year_2021
Mean: 1010.62
Standard Deviation: 19856.21
Minimum: 0.00
Maximum: 9141269.00

Sheet: Year_2022
Mean: 1132.77
Standard Deviation: 25543.65
Minimum: 0.00
Maximum: 5940246.00

Sheet: Year_2023
Mean: 1220.49
Standard Deviation: 19544.54
Minimum: 0.00
Maximum: 5711250.00

WLFR_PREMIUM_RCVD_AMT

Average Statistics for WLFR_PREMIUM_RCVD_AMT across all sheets:
Mean of Means: 2394116.44
Mean of Standard Deviations: 937505906.19
Overall Minimum: -130923183.00
Overall Maximum: 8701261226322.00

Statistics for each sheet:

Sheet: Year_1999

Mean: 15373687.25

Standard Deviation: 6507058397.56

Minimum: -23323344.00

Maximum: 3878411110812.00

Sheet: Year_2000

Mean: 940140.31

Standard Deviation: 655848416.88

Minimum: -30920892.00

Maximum: 501022020020.00

Sheet: Year_2001

Mean: 5305839.37

Standard Deviation: 3961101486.93

Minimum: -8201473.00

Maximum: 3000000000000.00

Sheet: Year_2002

Mean: 78997.47

Standard Deviation: 2485479.26

Minimum: -3844490.00

Maximum: 1127894523.00

Sheet: Year_2003

Mean: 72067.83

Standard Deviation: 1634604.80

Minimum: -5756324.00

Maximum: 367904006.00

Sheet: Year_2004

Mean: 16977056.25

Standard Deviation: 12116390706.69

Minimum: -130923183.00

Maximum: 8701261226322.00

Sheet: Year_2005

Mean: 93082.87

Standard Deviation: 3373281.40

Minimum: -6961535.00

Maximum: 1031570649.00

Sheet: Year_2006
Mean: 89277.43
Standard Deviation: 2426957.79
Minimum: -19645154.00
Maximum: 744821781.00

Sheet: Year_2007
Mean: 79508.49
Standard Deviation: 1844695.54
Minimum: -17170529.00
Maximum: 486492798.00

Sheet: Year_2008
Mean: 78493.40
Standard Deviation: 1929182.42
Minimum: -6356746.00
Maximum: 572542750.00

Sheet: Year_2009
Mean: 1208984.56
Standard Deviation: 8620140.86
Minimum: -5165871.00
Maximum: 683564496.00

Sheet: Year_2010
Mean: 1200608.79
Standard Deviation: 8150634.41
Minimum: -161261.00
Maximum: 643972812.00

Sheet: Year_2011
Mean: 1357688.14
Standard Deviation: 9375537.70
Minimum: -41744.00
Maximum: 672156213.00

Sheet: Year_2012
Mean: 1334077.88
Standard Deviation: 9003971.98
Minimum: -352730.00
Maximum: 713907036.00

Sheet: Year_2013

Mean: 1351143.04
Standard Deviation: 9074786.03
Minimum: -90958.00
Maximum: 770936207.00

Sheet: Year_2014
Mean: 1347478.53
Standard Deviation: 9123341.10
Minimum: -160530.00
Maximum: 799428326.00

Sheet: Year_2015
Mean: 1314744.21
Standard Deviation: 8977970.86
Minimum: -54885.00
Maximum: 891336610.00

Sheet: Year_2016
Mean: 1444899.01
Standard Deviation: 11623077.82
Minimum: -46753.00
Maximum: 1033138735.00

Sheet: Year_2017
Mean: 1371965.62
Standard Deviation: 10657481.20
Minimum: -527495.00
Maximum: 1169712787.00

Sheet: Year_2018
Mean: 1430535.32
Standard Deviation: 13264709.33
Minimum: -164745.00
Maximum: 1350541219.00

Sheet: Year_2019
Mean: 1484440.93
Standard Deviation: 17640843.29
Minimum: -26340.00
Maximum: 1976934319.00

Sheet: Year_2020
Mean: 1312724.96
Standard Deviation: 12210099.69

Minimum: -85607.00
Maximum: 1592661469.00

Sheet: Year_2021
Mean: 1369617.32
Standard Deviation: 14032207.82
Minimum: -42326.00
Maximum: 1849387058.00

Sheet: Year_2022
Mean: 1557656.12
Standard Deviation: 18581111.80
Minimum: -146485.00
Maximum: 2189318285.00

Sheet: Year_2023
Mean: 1678195.93
Standard Deviation: 23218531.60
Minimum: -442265.00
Maximum: 2453180903.00

WLFR_TOT_CHARGES_PAID_AMT

Average Statistics for WLFR_TOT_CHARGES_PAID_AMT across all sheets:

Mean of Means: 2900620.40
Mean of Standard Deviations: 1250374247.52
Overall Minimum: -50876783.00
Overall Maximum: 7708480105986.00

Statistics for each sheet:

Sheet: Year_1999
Mean: 2665821.87
Standard Deviation: 1124760034.13
Minimum: -1364646.00
Maximum: 511150282545.00

Sheet: Year_2000
Mean: 213424.50
Standard Deviation: 19940838.54
Minimum: -8289273.00
Maximum: 10917007917.00

Sheet: Year_2001
Mean: 904892.43
Standard Deviation: 441396713.48
Minimum: -1315732.00
Maximum: 328563658510.00

Sheet: Year_2002
Mean: 20858868.64
Standard Deviation: 10873961269.16
Minimum: -50876783.00
Maximum: 571923333333.00

Sheet: Year_2003
Mean: 2112857.35
Standard Deviation: 1406478836.90
Minimum: -8167154.00
Maximum: 1029138029133.00

Sheet: Year_2004
Mean: 226368.61
Standard Deviation: 5428072.56
Minimum: -2373277.00
Maximum: 2691621021.00

Sheet: Year_2005
Mean: 256561.94
Standard Deviation: 16033622.44
Minimum: -6961535.00
Maximum: 10358710356.00

Sheet: Year_2006
Mean: 239416.75
Standard Deviation: 4078612.09
Minimum: -976350.00
Maximum: 966537000.00

Sheet: Year_2007
Mean: 245567.25
Standard Deviation: 4498055.91
Minimum: -2795760.00
Maximum: 1234812800.00

Sheet: Year_2008
Mean: 228099.84

Standard Deviation: 2878510.96
Minimum: -18423904.00
Maximum: 887691107.00

Sheet: Year_2009
Mean: 572107.59
Standard Deviation: 6667240.88
Minimum: -426428.00
Maximum: 2279234232.00

Sheet: Year_2010
Mean: 562189.11
Standard Deviation: 5249304.96
Minimum: -484204.00
Maximum: 1619767769.00

Sheet: Year_2011
Mean: 581563.35
Standard Deviation: 6524144.16
Minimum: -758096.00
Maximum: 1684131637.00

Sheet: Year_2012
Mean: 612977.82
Standard Deviation: 21571525.88
Minimum: -338488.00
Maximum: 9482987651.00

Sheet: Year_2013
Mean: 559397.07
Standard Deviation: 3820717.04
Minimum: -1132735.00
Maximum: 596001549.00

Sheet: Year_2014
Mean: 586686.41
Standard Deviation: 8783507.31
Minimum: -1379539.00
Maximum: 3526873000.00

Sheet: Year_2015
Mean: 633030.31
Standard Deviation: 27968407.11
Minimum: -473154.00

Maximum: 13113073061.00

Sheet: Year_2016

Mean: 33550137.00

Standard Deviation: 15944385732.65

Minimum: -990278.00

Maximum: 7708480105986.00

Sheet: Year_2017

Mean: 578324.93

Standard Deviation: 5596131.21

Minimum: -417068.00

Maximum: 1786974328.00

Sheet: Year_2018

Mean: 599690.20

Standard Deviation: 10347967.91

Minimum: -9542211.00

Maximum: 4458298679.00

Sheet: Year_2019

Mean: 3176789.63

Standard Deviation: 1289111480.78

Minimum: -697801.00

Maximum: 644491643655.00

Sheet: Year_2020

Mean: 597155.34

Standard Deviation: 5034535.64

Minimum: -3788857.00

Maximum: 901602564.00

Sheet: Year_2021

Mean: 613567.26

Standard Deviation: 6857566.79

Minimum: -562736.00

Maximum: 1643106327.00

Sheet: Year_2022

Mean: 638722.01

Standard Deviation: 6920032.52

Minimum: -2990211.00

Maximum: 1807949739.00

Sheet: Year_2023
Mean: 701292.74
Standard Deviation: 11063326.93
Minimum: -1553909.00
Maximum: 4540044555.00

WLFR_PREMIUM_RCVD_AMT_PER_CAPITA

Average Statistics for WLFR_PREMIUM_RCVD_AMT_PER_CAPITA across all sheets:

Mean of Means: 21864.86
Mean of Standard Deviations: 13302903.79
Overall Minimum: -178856.81
Overall Maximum: 204126900569.05

Statistics for each sheet:

Sheet: Year_1999
Mean: 497355.78
Standard Deviation: 317606747.54
Minimum: -167793.84
Maximum: 204126900569.05

Sheet: Year_2000
Mean: 939.07
Standard Deviation: 541780.03
Minimum: -5469.90
Maximum: 413725862.94

Sheet: Year_2001
Mean: 359.62
Standard Deviation: 106816.38
Minimum: -14029.00
Maximum: 79611495.90

Sheet: Year_2002
Mean: 194.45
Standard Deviation: 14785.44
Minimum: -7002.71
Maximum: 7510664.00

Sheet: Year_2003
Mean: 325.31
Standard Deviation: 133072.64
Minimum: -10409.27

Maximum: 97302682.00

Sheet: Year_2004

Mean: 19583.29

Standard Deviation: 13926898.04

Minimum: -178856.81

Maximum: 10001449685.43

Sheet: Year_2005

Mean: 168.18

Standard Deviation: 8377.70

Minimum: -4420.58

Maximum: 4864419.26

Sheet: Year_2006

Mean: 141.19

Standard Deviation: 4452.28

Minimum: -8692.55

Maximum: 1158915.50

Sheet: Year_2007

Mean: 121.13

Standard Deviation: 2310.33

Minimum: -12160.43

Maximum: 719315.67

Sheet: Year_2008

Mean: 165.78

Standard Deviation: 19261.62

Minimum: -16906.24

Maximum: 9362321.65

Sheet: Year_2009

Mean: 1425.80

Standard Deviation: 24341.72

Minimum: -7756.56

Maximum: 4557096.64

Sheet: Year_2010

Mean: 1257.54

Standard Deviation: 8262.34

Minimum: -15493.00

Maximum: 899956.93

Sheet: Year_2011
Mean: 1596.44
Standard Deviation: 24915.15
Minimum: -965.22
Maximum: 4426571.00

Sheet: Year_2012
Mean: 1544.61
Standard Deviation: 8423.88
Minimum: -9555.00
Maximum: 930275.37

Sheet: Year_2013
Mean: 2014.55
Standard Deviation: 48778.85
Minimum: -4040.00
Maximum: 8981868.57

Sheet: Year_2014
Mean: 1742.20
Standard Deviation: 6340.56
Minimum: -6979.57
Maximum: 439823.59

Sheet: Year_2015
Mean: 1776.61
Standard Deviation: 8188.22
Minimum: -3127.33
Maximum: 656513.92

Sheet: Year_2016
Mean: 1915.42
Standard Deviation: 9067.84
Minimum: -324.67
Maximum: 1103780.75

Sheet: Year_2017
Mean: 1850.59
Standard Deviation: 7464.64
Minimum: -4508.50
Maximum: 874030.45

Sheet: Year_2018
Mean: 1896.44

Standard Deviation: 6763.25
Minimum: -125580.00
Maximum: 850643.95

Sheet: Year_2019
Mean: 1995.18
Standard Deviation: 9843.62
Minimum: -26340.00
Maximum: 882325.70

Sheet: Year_2020
Mean: 1993.55
Standard Deviation: 7876.00
Minimum: -216.33
Maximum: 806761.45

Sheet: Year_2021
Mean: 1991.22
Standard Deviation: 10507.68
Minimum: -661.34
Maximum: 1153207.96

Sheet: Year_2022
Mean: 2038.80
Standard Deviation: 9124.72
Minimum: -180.04
Maximum: 913900.46

Sheet: Year_2023
Mean: 2228.83
Standard Deviation: 18194.21
Minimum: -0.01
Maximum: 1610604.71

WLFR_TOT_CHARGES_PAID_AMT_PER_CAPITA

Average Statistics for WLFR_TOT_CHARGES_PAID_AMT_PER_CAPITA across all sheets:
Mean of Means: 33723.96
Mean of Standard Deviations: 16920728.97
Overall Minimum: -4771105.50
Overall Maximum: 211823456790.11

Statistics for each sheet:

Sheet: Year_1999
Mean: 28918.82
Standard Deviation: 10398198.55
Minimum: -75475.00
Maximum: 3901910553.78

Sheet: Year_2000
Mean: 1527.29
Standard Deviation: 391694.24
Minimum: -68467.00
Maximum: 296374392.31

Sheet: Year_2001
Mean: 2016.48
Standard Deviation: 565370.25
Minimum: -27707.17
Maximum: 326604034.30

Sheet: Year_2002
Mean: 766527.49
Standard Deviation: 402739308.14
Minimum: -656676.00
Maximum: 211823456790.11

Sheet: Year_2003
Mean: 9099.69
Standard Deviation: 6141702.06
Minimum: -108224.35
Maximum: 4494052528.97

Sheet: Year_2004
Mean: 897.42
Standard Deviation: 19440.03
Minimum: -254675.00
Maximum: 5450314.49

Sheet: Year_2005
Mean: 966.36
Standard Deviation: 64025.14
Minimum: -185834.50
Maximum: 43341884.33

Sheet: Year_2006
Mean: 844.02

Standard Deviation: 12238.34
Minimum: -106317.00
Maximum: 3702753.00

Sheet: Year_2007
Mean: 994.66
Standard Deviation: 59988.12
Minimum: -10471.01
Maximum: 29369825.00

Sheet: Year_2008
Mean: 793.52
Standard Deviation: 8165.67
Minimum: -146221.46
Maximum: 2496345.00

Sheet: Year_2009
Mean: 1711.31
Standard Deviation: 9164.84
Minimum: -35961.00
Maximum: 1075121.00

Sheet: Year_2010
Mean: 1691.22
Standard Deviation: 13390.94
Minimum: -6673.00
Maximum: 3757527.00

Sheet: Year_2011
Mean: 1739.36
Standard Deviation: 16902.01
Minimum: -61132.50
Maximum: 5851544.00

Sheet: Year_2012
Mean: 1661.71
Standard Deviation: 13166.47
Minimum: -9905.00
Maximum: 4019209.50

Sheet: Year_2013
Mean: 1626.85
Standard Deviation: 8861.45
Minimum: -28719.86

Maximum: 1305290.75

Sheet: Year_2014

Mean: 1705.86

Standard Deviation: 22300.12

Minimum: -30196.00

Maximum: 7201137.77

Sheet: Year_2015

Mean: 1876.23

Standard Deviation: 99118.41

Minimum: -3240.50

Maximum: 46832403.79

Sheet: Year_2016

Mean: 1663.16

Standard Deviation: 19051.76

Minimum: -13632.80

Maximum: 6417519.00

Sheet: Year_2017

Mean: 1626.99

Standard Deviation: 15425.41

Minimum: -15125.00

Maximum: 5019590.81

Sheet: Year_2018

Mean: 1650.91

Standard Deviation: 16690.08

Minimum: -4771105.50

Maximum: 2444098.00

Sheet: Year_2019

Mean: 6335.65

Standard Deviation: 2302570.51

Minimum: -100880.00

Maximum: 1169676304.27

Sheet: Year_2020

Mean: 1776.16

Standard Deviation: 15789.80

Minimum: -131342.00

Maximum: 6705783.50

Sheet: Year_2021

Mean: 1756.14

Standard Deviation: 24390.53

Minimum: -60235.00

Maximum: 10018941.02

Sheet: Year_2022

Mean: 1807.57

Standard Deviation: 17823.79

Minimum: -34137.00

Maximum: 7822642.00

Sheet: Year_2023

Mean: 1884.08

Standard Deviation: 23447.55

Minimum: -19126.00

Maximum: 7394209.37

Appendix Potential Additions / Next steps

Adding to our equation

α_s - State fixed effect

δ_t - year fixed effects

New:

$$Y_{ist} = \beta_0 + \beta_1 (ACA_s * Post_t) + \beta_2 ACA_s + \beta_3 Post_t + \gamma Controls + \alpha_s + \delta_t + \varepsilon_{ist}$$

β_2 captures the time effect after ACA implementation

- Add to interpretation maybe

Control Variable Descriptive Stats

Log transform Y variable and its effect on interpretation